

NEWS

European Water Chestnut Range Expands to 15 Pennsylvania Counties

This invasive aquatic plant was previously reported in only eleven northwestern and eastern Pennsylvania counties.

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European Water Chestnut (*Trapa natans*) is a rooted, floating aquatic plant that is native to Europe, Asia, and Africa. It was first observed in North America in the late 1800s where it was intentionally being grown in a botanical garden at Harvard University and later escaped into the wild. This aquatic invader has now spread throughout New England and the Mid-Atlantic Region. As of November 2020, this species had only been reported in eleven Pennsylvania counties with most of the infestations located in

southeastern Pennsylvania in Bucks and Montgomery Counties. Other counties with limited or isolated occurrences included Berks, Chester, Northampton, Luzerne, Pike, Susquehanna, Wayne, Warren, and Mercer counties. In 2023, an isolated population was reported in a farm pond in northern Indiana County for the first time, and as of May 2025, the range has expanded to include Dauphin, Monroe, and Philadelphia counties as well (Figure 2).

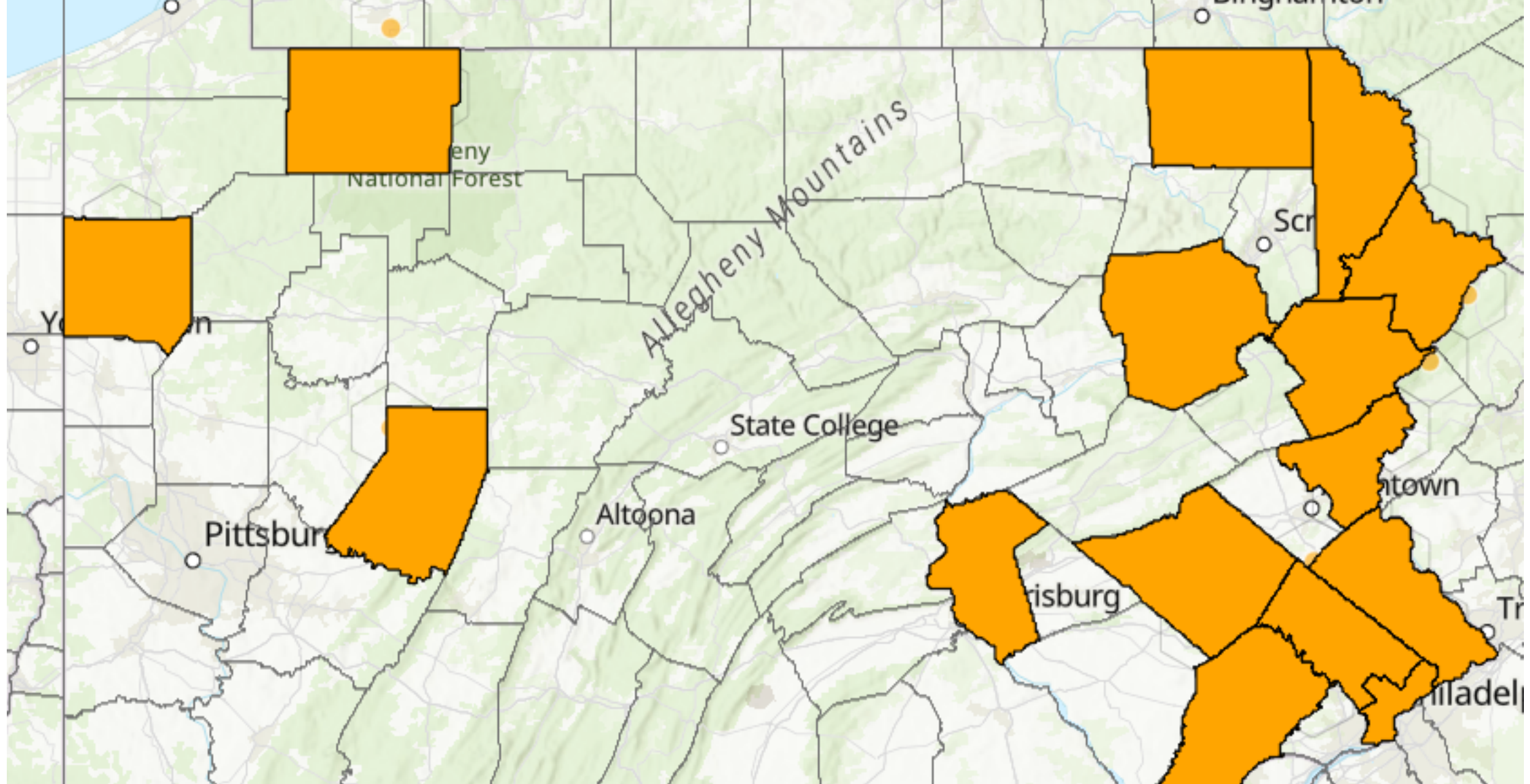


Figure 2. Reported distribution of European Water Chestnut across Pennsylvania ([iMapInvasives](#): NatureServe's online data system supporting strategic invasive species management. ©2025, NatureServe. (Date accessed: 05/5/2025))

Identifying European Water Chestnut is fairly simple because of its unique and unmistakable characteristics. The floating leaves are green, glossy, and roundly-triangular with toothed-edges that form a rosette on the water's surface (Figure 3). It has small, white flowers that grow individually at the center of the stem, typically appearing in mid to late July. The fruit forms about one month later, a large seed or nut with four short spines that are sharp enough to puncture shoes (Figure 4). No other aquatic plants are considered to be a look-a-like.

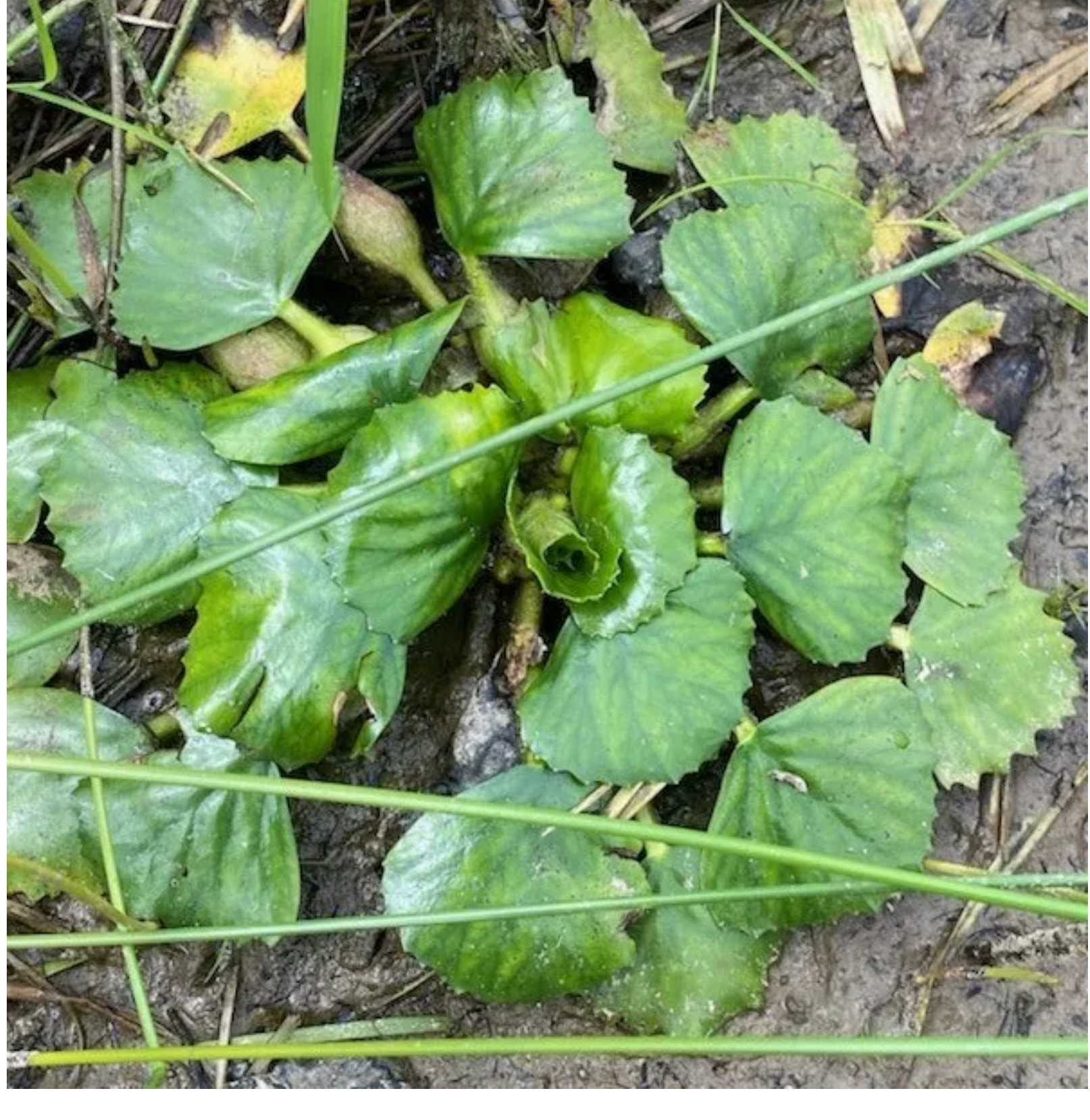


Figure 3. The leaves of the European Water Chestnut are serrated and grow in a rosette pattern (Danielle Rhea, Penn State).



Figure 4. The seed pods of European Water Chestnuts can be easily seen on the underside of the rosette and have four distinctive spikes (Danielle Rhea, Penn State).

As an annual species, the primary mode of reproduction is by seeds. An individual seed can produce up to 10 to 15 rosettes which can then produce up to 20 seeds each. Seeds can overwinter in Pennsylvania and will sometimes remain viable for up to 12 years. European Water Chestnut can spread both by seeds being moved from place to place on other objects, birds, or animals as well as by fragmentation if rosettes are detached from their stem and allowed to float or are moved to another area. This plant can grow in nearly any freshwater body but prefers shallow, nutrient-rich ponds and lakes or even slow-moving streams and rivers with soft, muddy bottoms. The cord-like stem of this plant can grow up to approximately 15 feet, which means it can completely cover many ponds in Pennsylvania once established. As it spreads, it forms dense mats that can interfere with recreation as well as displace native species and shade out submerged plants, leading to declines in dissolved oxygen (Figure 1).

Considered a Class A Noxious Weed, populations of European Water Chestnut should be reported and eradicated as quickly as possible. In Pennsylvania, aquatic invasive species should be reported to the Pennsylvania Fish and Boat Commission. This can be done with an [online form](#), by calling in a voicemail report to 1-833-INVASIV, or by using the [Pennsylvania Aquatic Invasive Species App](#) on Apple and Android devices. Invasive species can also be reported through [PA iMapInvasives](#).

When it comes to eliminating invasive populations of European Water Chestnut, the sooner action is taken, the better; however, it is helpful to keep in mind its reproductive strategy to optimize your effort. Since the primary mode of reproduction is through seeds, the best timing of eradication efforts is in the late spring or early summer – after much of the plant growth has occurred but before seeds are released. Any actions taken to control the plant after seed dispersal has occurred will have very little or no impact since any future growth is dependent on seeds germinating the following year.

There are both physical and chemical options for controlling this plant. Physical removal includes hand-pulling by boat or the use of mechanical harvesters. For early detected, small populations, hand-pulling is highly effective. With physical removal, take care to remove all displaced rosettes since this plant can spread by fragmentation. Additionally, dispose of all plant material far away from water, ideally in a sealed trash bag. Chemical control options include aquatic herbicides with the active ingredient 2,4-D or triclopyr. For larger infestations, a control approach combining both mechanical control and the use of an herbicide may be needed. Additionally, because of the long-term viability of the seeds, control and monitoring of impacted waterbodies may be needed for many years. Keep in mind that the application of aquatic herbicides on any Pennsylvania waterbody, including private ponds, does require a permit from the Pennsylvania Department of Environmental Protection. For more information on how to complete this permit, consult our [Guide for Completing the State Application for Aquatic Herbicide Use](#).

Doing our best to prevent European Water Chestnut and all aquatic invasive species from spreading will always be much easier than eradicating an invasive population after it is established. While some spread will be out of our control, such as movement by wildlife, following the mantra, "clean, drain, dry" after spending time in and around the water will help slow the spread of invasive species. More specifically, clean all plant material and mud from boats and gear; drain all equipment, buckets, and live wells on boats; and disinfect or dry boots, gear, boats, and trailers for at least five days before spending time in another waterbody. For more information, consult the article, "[Slow the Spread of Aquatic Invasive Species](#)."

Sources and Additional Information

- Golden Sands Resource Conservation & Development Council, Inc. "[Aquatic Invasive Species Quick Guide Water Chestnut \(*Trapa natans* L.\)](#)." Accessed June 2024.
- Penn State Extension (2022). "[Focus on Aquatic Invasive Species – European Water Chestnut](#)." Accessed May 2024.
- Pennsylvania Department of Conservation and Natural Resources. "[Invasive Plants in Pennsylvania European water chestnut](#)." Accessed May 2024.
- Pennsylvania Fish and Boat Commission (2020). "[Aquatic Invasive Species \(AIS\) Control Plan: European Water Chestnut](#)." Accessed May 2024.
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Authors

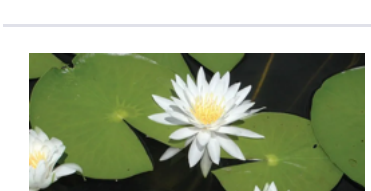
Danielle Rhea
Extension Educator

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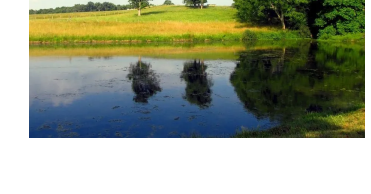
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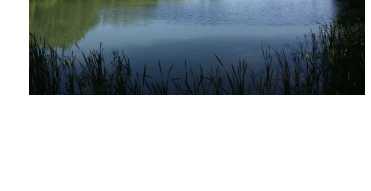
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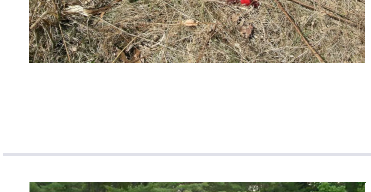
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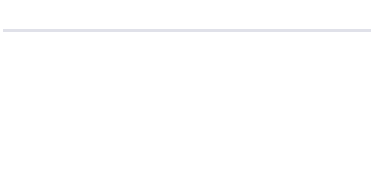
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